

Supervised Machine Learning for Diabetic Retinopathy Detection: SVM and Random Forest Approaches

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Background, Motivation, and Significance

Diabetic Retinopathy is a serious condition caused by diabetes. High blood sugar damages tiny blood vessels in the retina, potentially leading to blindness if not diagnosed early in this diabetic condition. Doctors manually examine eye images to detect this condition which can be very slow, expensive, and higher chances of leading to human error. Automatic detection using machine learning might help quickly and accurately in diagnosing Diabetic Retinopathy. In this project we aim to develop reliable machine learning methods to assist doctors in making DR diagnosis faster, cheaper, and accurate.

Research Questions and Hypotheses

1. Do Support Vector Machines (SVM) and Random Forest machine learning models achieve successful detection of diabetic retinopathy from retinal image analysis?

Hypothesis: Both machine learning techniques will detect diabetic retinopathy effectively although their accuracy levels might show minor differences.

2. Does SVM or Random Forest demonstrate superior accuracy when it comes to detecting diabetic condition?

Hypothesis: Random Forest is usually strong, but in our case, SVM did better. This is because SVM works well with high-dimensional data and smaller sample sizes.

3. What impact do preprocessing steps like grayscale conversion and histogram equalization on model accuracy?

Hypothesis: The preprocessing of images improves diagnostic model accuracy by making essential details clear.

Data Set

Our study utilized the DIARETDB0 dataset which comprises 130 retinal images. The dataset includes 110 retinal images that display diabetic retinopathy and 20 images without any disease. Every image received a distinct label where '1' indicated diabetic retinopathy and '0' represented a healthy retina. We used raw

pixel intensity values. We also looked at grouped intensity areas and features from wavelet and filtering methods. The binary DR label serving as the target variable took on values of either 1 or 0.

Metric	Count
Total Images	130
DR-Positive Images	110
Healthy Images	20
Image Size	200x200 pixels

Table 1: Summary of Dataset Characteristics

Methodology

Data Preparation:

We performed extensive preprocessing to guarantee superior quality input for our machine learning models.

- **Image Resizing:** We standardized all images by resizing them to 200x200 pixels.
- **Grayscale Conversion:** The transformation of color images into grayscale format helped decrease processing complexity.
- **Histogram Equalization:** This method enhanced image contrast which allowed important features such as blood vessels and lesions to stand out more clearly. These equalized images were then passed through DWT, matched filters, and Gabor filters for further enhancement.
- **Discrete Wavelet Transform (DWT & IDWT):** The Discrete Wavelet Transform (DWT & IDWT) method functions to enhance important details and suppress image noise.
- **Matched and Gabor Filters:** Matched filters helped to show blood vessels while Gabor filters emphasized textures and fine details essential for detecting DR.

No missing or corrupted images were found. The dataset underwent identical preprocessing procedures throughout its entirety.



Fig 1: Original Retinal Image



Fig 2: Histogram Equalized Image



Fig 3: Image After Wavelet and Filtering

Feature Selection and Transformation:

To reduce complexity, we used:

- **K-means Clustering:** This algorithm sorted image pixels into three groups based on intensity. This helped identify important areas like lesions and vessels.
- **Principal Component Analysis (PCA):** We decreased data dimensionality from roughly 40,000 features to 10 principal components to improve computational speed and prevent model overfitting.

Class Imbalance Handling:

After our initial model evaluation and presentation, we realized that the dataset was highly imbalanced. Dataset has significantly fewer healthy images than DR-positive ones. So we have to implement SMOTE (Synthetic Minority Oversampling Technique). This method synthetically generates set of new samples for the minority class. This adjustment was made post-presentation to improve model fairness and robustness.

Machine Learning Models and Validation:

- **Support Vector Machine (SVM):** We selected SVM because of its excellent classification performance which proves beneficial for smaller datasets. The RBF kernel functioned with hyperparameters $C=10$ and $\gamma=0.01$ to create effective distinctions between DR images and healthy images.
- **Random Forest:** The model was selected because it handles complex datasets well and at the same time minimizing overfitting risks. This model used 100 trees as estimators with a maximum depth of 5.

We used a set split. The training and testing images were listed in separate text files that came with the dataset. This ensured consistency with the dataset provider's setup.

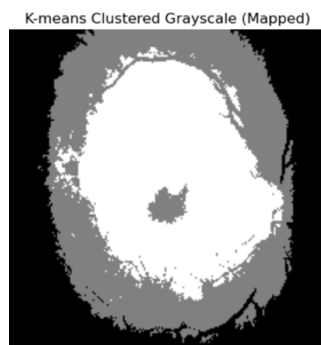


Fig 4: Image Segmented by K-means Clustering

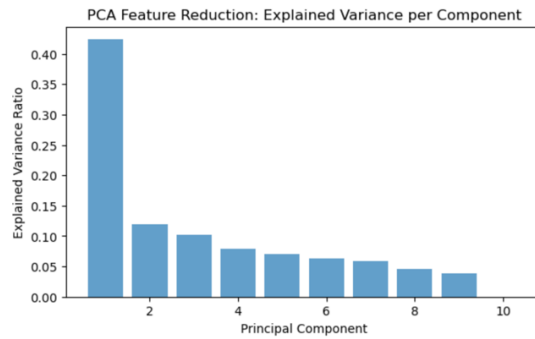


Fig 5: PCA Feature Reduction Plot

Results

- **Support Vector Machine Accuracy: 85%**
- **Random Forest Accuracy: 68%**

SVM slightly outperformed Random Forest. This outcome may be attributed to SVM's efficiency with high-dimensional but limited data. The results also confirmed that preprocessing techniques significantly enhanced model performance by improving feature clarity and reducing noise.

Model	Accuracy
SVM	85%
Random Forest	68%

Table 2: Accuracy Comparison

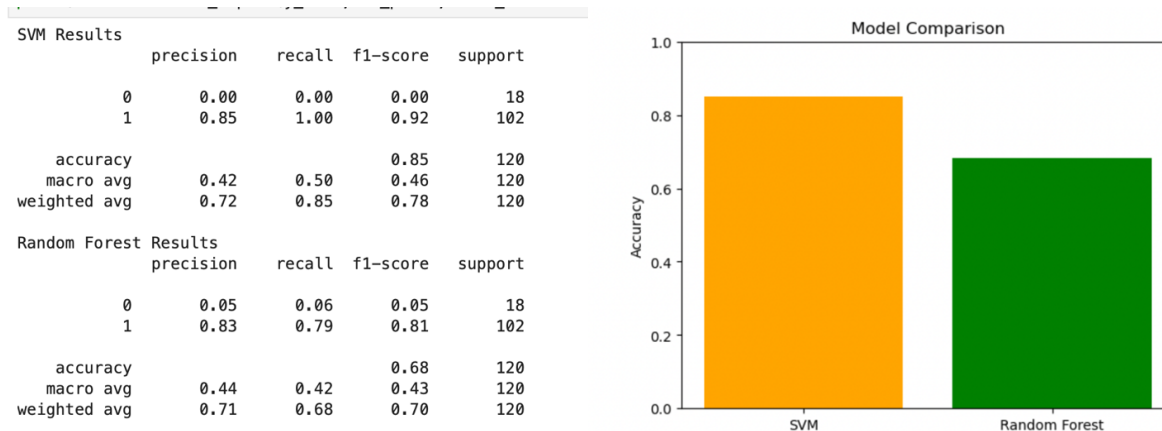


Fig 6: Results and Accuracy Comparison Bar Chart

Discussion

Machine learning methods shown success in identifying diabetic retinopathy according to study findings. Methods like SVM can reliably find diabetic retinopathy. Random Forest may need more tuning for better results. The preprocessing and feature transformation steps produced clearer features which had a substantial positive effect on the models' accuracy.

Conclusion

The study demonstrates that automated machine learning algorithms such as SVM and Random Forest provide dependable detection of diabetic retinopathy in retinal images. The developed methods can significantly support healthcare providers through faster diagnostics and reduced human mistakes while enabling prompt treatment which prevents blindness and improves patient results.